

Q9(a) asks "Which of these could be the output for a dynamo?" - This is about generators/dynamos

Q9(b)(i) asks about transformer not working with DC input

Q9(b)(ii) transformer calculation

Q9(c) about high voltage transmission using transformers

Q5 has questions about magnetic fields and electromagnetic forces on wires

Q6 about electromagnetic induction with bar magnet, coil, and voltmeter

Q6(a)(i) "Draw a diagram to show how the teacher should arrange the apparatus" for electromagnetic induction

Q6(a)(ii) "Explain how the teacher could use this apparatus to demonstrate the factors affecting the size and direction of the induced potential difference" - This directly matches 13.2!

Q6(b) about transformers

Q7 has questions about electromagnetic induction with magnet and coil

Q7(a)(i) asks about deflection on milliammeter when magnet moves into/out of coil - directly relates to 13.1 and 13.2

Q7(a)(ii) asks about ways to increase deflections - relates to factors in 13.2

Q10(a) about magnet and coil with sensitive centre-zero ammeter - directly relates to 13.1

Q10(a) "Explain what will be observed on the meter when the magnet is pushed in and pulled out of the coil, repeatedly" - This matches 13.1P and 13.2!

Q10(b) about transformer calculations

Q10(c) about electricity delivery from power station to homes

Q10 has questions about a model - still checking

Q10(a) "Explain why the voltmeter shows a reading as the magnet passes through the coil" - This matches 13.1P and 13.2!

Q10(b) transformer explanation

Q10(a)(ii) describes how to investigate how the reading depends on height the magnet is dropped from - relates to 13.1 and 13.2

Q9(c) about coil rotating between poles of magnet and how current changes - this is about a.c. generators/alternators, matching 13.3P!

Q9(a) about transformer currents

Q9(c) asks to "Explain why the current changes in the way shown by the graph... include details of the position of the coil relative to the magnet at each of the times labelled P, Q, R, S and T" - This relates to

alternators generating a.c. (13.3P)

From **Jun-2019, Q6(a)(ii) explicitly asks about "factors affecting the size and direction of the induced potential difference" - this directly matches 13.2.**

From **Jun-2020, Q7(a)(ii) asks about increasing deflections on the milliammeter, relating to factors in 13.2.**

From **Jun-2022, Q10(a) about observing meter readings when magnet is pushed in and out directly matches 13.1P and 13.2.**

From **Jun-2024, Q10(a) explains why voltmeter shows reading as magnet passes through coil, matching 13.1P and 13.2.**

From **Nov-2021, Q9(c) about coil rotating between magnet poles and how current changes relates to alternators and a.c. generation (13.3P), though it doesn't explicitly mention "alternator."**

From jun-2019 | Page 16:

Q6(a)(ii): "Explain how the teacher could use this apparatus to demonstrate the factors affecting the size and direction of the induced potential difference."

From jun-2020 | Page 22:

Q7(a)(i): "Which row of the table shows the deflection on the milliammeter for the three stages in Figure 15?" (magnet moving into coil, stationary inside, magnet moving out)

From jun-2022 | Page 30:

Q10(a): "Explain what will be observed on the meter when the magnet is pushed in and pulled out of the coil, repeatedly."

From jun-2024 | Page 32:

Q10(a): "Explain why the voltmeter shows a reading as the magnet passes through the coil."

From nov-2021 | Page 28:

Q9(c): "Explain why the current changes in the way shown by the graph in Figure 20. Your answer should include details of the position of the coil relative to the magnet at each of the times labelled P, Q, R, S and T."